# Derivatives Using the Power Rule

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Name: _____ Date: _____

**Learning Objectives**

- Apply the Power Rule to find derivatives of polynomial and rational functions
- Convert radical and rational expressions into exponential form before differentiating
- Simplify derivative results using algebraic manipulation and negative/fractional exponents

## Problems

**1.** Find the derivative of the function using the Power Rule.

$$y = x^5$$

**2.** Find the derivative of the function using the Power Rule.

$$y = 4x^3 - 7x^2 + 2x - 9$$

**3.** Find the derivative of the function. Rewrite the fraction by separating constants before applying the Power Rule.

$$y = \frac{x^4}{4} + \frac{x^3}{3} - x$$

**4.** Convert the square root to a fractional exponent, then find the derivative.

$$f(x) = \sqrt{x}$$

**5.** Convert the cube root to a fractional exponent, then find the derivative.

$$g(x) = 3\sqrt[3]{x^2}$$



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**6.** Rewrite the expression by dividing each term in the numerator by x, then find the derivative using the Power Rule.

$$f(x) = \frac{x^3 + 5x^2 - 2}{x}$$

**7.** Find the derivative of the function, which includes an exponential term. Recall that the derivative of e to the x equals e to the x.

$$h(x) = 6x^4 - 5e^x + 3$$

**8.** Convert all radical and negative-power terms into fractional exponents, then find the derivative and simplify.

$$g(x) = \frac{5}{x^{3/5}} - 7e^x - 4\sqrt[3]{x}$$

**9.** Rewrite the function by distributing x to the negative 1 to eliminate the denominator, then find the derivative.

$$f(x) = \frac{x^3 - 4x + 6}{x^2}$$

**10.** Find the derivative of the function by first rewriting all terms using positive or negative integer and fractional exponents, then apply the Power Rule to each term and simplify fully.

$$y = \frac{x^3}{3} + \frac{x^2}{2} + x - \frac{3}{\sqrt{x}} + 2e^x$$

# Derivatives Using the Power Rule — Answer Key

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## Answer Key

**1. Answer: y' = 5x^4**

- Apply the Power Rule: if y = x^n, then y' = n·x^(n–1)
- Bring down the exponent: y' = 5·x^(5–1)
- Simplify: y' = 5x^4

**2. Answer: y' = 12x^2 - 14x + 2**

- Differentiate each term separately using the Power Rule
- d/dx(4x^3) = 12x^2
- d/dx(–7x^2) = –14x
- d/dx(2x) = 2
- d/dx(–9) = 0 (constant rule)
- Combine: y' = 12x^2 – 14x + 2

**3. Answer: y' = x^3 + x^2 - 1**

- Rewrite as y = (1/4)x^4 + (1/3)x^3 – x
- d/dx[(1/4)x^4] = (1/4)·4x^3 = x^3
- d/dx[(1/3)x^3] = (1/3)·3x^2 = x^2
- d/dx[–x] = –1
- Combine: y' = x^3 + x^2 – 1

**4. Answer: f'(x) = 1/(2√x) or (1/2)x^(–1/2)**

- Rewrite: f(x) = x^(1/2)
- Apply the Power Rule: f'(x) = (1/2)x^(1/2 – 1)
- Simplify the exponent: 1/2 – 1 = –1/2
- f'(x) = (1/2)x^(–1/2) = 1/(2√x)

**5. Answer: g'(x) = 2x^(–1/3) or 2/∛x**

- Rewrite: g(x) = 3x^(2/3)
- Apply the Power Rule: g'(x) = 3·(2/3)x^(2/3 – 1)
- Simplify the coefficient: 3·(2/3) = 2
- Simplify the exponent: 2/3 – 1 = 2/3 – 3/3 = –1/3
- g'(x) = 2x^(–1/3) = 2/∛x

**6. Answer: f'(x) = 2x + 5 + 2x^(–2)**

- Divide each term by x: f(x) = x^2 + 5x – 2x^(–1)
- Apply the Power Rule to each term
- d/dx[x^2] = 2x
- d/dx[5x] = 5
- d/dx[–2x^(–1)] = –2·(–1)x^(–2) = 2x^(–2)

- Combine: f'(x) = 2x + 5 + 2x^(–2)

**7. Answer: h'(x) = 24x^3 - 5e^x**

- d/dx[6x^4] = 24x^3
- d/dx[–5e^x] = –5e^x (since d/dx[e^x] = e^x)
- d/dx[3] = 0
- Combine: h'(x) = 24x^3 – 5e^x

**8. Answer: g'(x) = –3x^(–8/5) – 7e^x – (4/3)x^(–2/3)**

- Rewrite: g(x) = 5x^(–3/5) – 7e^x – 4x^(1/3)
- d/dx[5x^(–3/5)] = 5·(–3/5)x^(–3/5 – 1) = –3x^(–8/5)
- d/dx[–7e^x] = –7e^x
- d/dx[–4x^(1/3)] = –4·(1/3)x^(1/3 – 1) = –(4/3)x^(–2/3)
- Combine: g'(x) = –3x^(–8/5) – 7e^x – (4/3)x^(–2/3)

**9. Answer: f'(x) = 1 + 4x^(–3) – 12x^(–4)**

- Divide each term by x^2: f(x) = x – 4x^(–1) + 6x^(–2)
- d/dx[x] = 1
- d/dx[–4x^(–1)] = –4·(–1)x^(–2) = 4x^(–2)
- Wait — recheck: d/dx[–4x^(–1)] = 4x^(–2)
- d/dx[6x^(–2)] = 6·(–2)x^(–3) = –12x^(–3)
- Combine: f'(x) = 1 + 4x^(–2) – 12x^(–3)

**10. Answer: y' = x^2 + x + 1 + (3/2)x^(–3/2) + 2e^x**

- Rewrite: y = (1/3)x^3 + (1/2)x^2 + x – 3x^(–1/2) + 2e^x
- d/dx[(1/3)x^3] = (1/3)·3x^2 = x^2
- d/dx[(1/2)x^2] = (1/2)·2x = x
- d/dx[x] = 1
- d/dx[–3x^(–1/2)] = –3·(–1/2)x^(–3/2) = (3/2)x^(–3/2)
- d/dx[2e^x] = 2e^x
- Combine: y' = x^2 + x + 1 + (3/2)x^(–3/2) + 2e^x